LEARNING TO IGNORE WHAT MATTERS: A JEPA WORLD MODEL KEEPS THE CUBE AND LOSES THE WALL

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ABSTRACT

A joint-embedding predictive world model (JEPA) learns its latent by predicting its own next latent and keeping the latent marginal Gaussian, with no decoder and no reward. We ask what such a model keeps when survival depends on one thing: a moving wall whose gap is revealed late. The JEPA imagines a cube in an empty room. Its encoder keeps the wall: an off-budget decoder renders it from real encodings, and a convolutional policy trained on those decoded frames learns the retreat-and-wait strategy in two of nine runs, as it does on true pixels in two of three. Its imagination loses it: with the same MLP torso, PPO trained inside the JEPA passes 0.27 walls per episode over 27 runs, against 0.94 when trained in the real gate on the JEPA's own frozen encoders. The deficit belongs to the imagination pipeline as a whole, predictor, fitted reward and termination heads and PPO protocol together; sampling the fitted residual predictor does not remove it. A flow model trained on the same data and budget keeps the wall. We prove that the objective permits this: a non-collapsed global optimum can carry no state at all, and an exactly Gaussian, fully decodable latent can hold a control-relevant feature at arbitrarily small amplitude. Non-collapse does not guarantee control-relevant state, and decodable state does not guarantee that imagination uses it.

1 INTRODUCTION

A world model can retain observable state without supporting a policy trained in imagination. We study this distinction in a JEPA trained with squared next-latent prediction and a Gaussian marginal regularizer, without reconstruction or reward supervision during representation learning (Maes et al., 2026; Balestriero & LeCun, 2025). Non-collapse constrains the distribution of features; it does not specify which features matter for control or how accurately imagination must preserve them.

Our gate task makes this distinction observable. A moving wall approaches a controlled cube, but reveals its gap too late for an agent waiting at the centre to respond. Survival requires retreating before the reveal and then moving toward the gap. Real frames encoded and decoded through the frozen JEPA retain the wall; decoded predicted rollouts lose it (Figure 1).

The primary comparison holds the frozen encoders and MLP torso design fixed. Policies trained on real transitions pass 0.94 walls per episode, versus 0.27 for policies trained inside the JEPA. The latter use learned transitions, rewards and terminations, and a different PPO collection and update protocol. The contrast measures the resulting pipeline deficit; it does not locate its cause. The same MLP design on true pixels also fails to discover the strategy.

A convolutional policy on decoded frozen latents discovers the strategy in $2/9$ runs, both on encoder 1. At least one fitted representation thus supports successful control with decoder supervision. A single flow-model checkpoint supplies a feasibility reference under the same data and model-training budget, without establishing a ranking between model families.

Contributions. We establish a measured deficit of the tested imagination-and-training pipeline, an existence result for useful information in its frozen representation, and population counterexamples to control sufficiency under its objective. The counterexamples separate Gaussian non-collapse

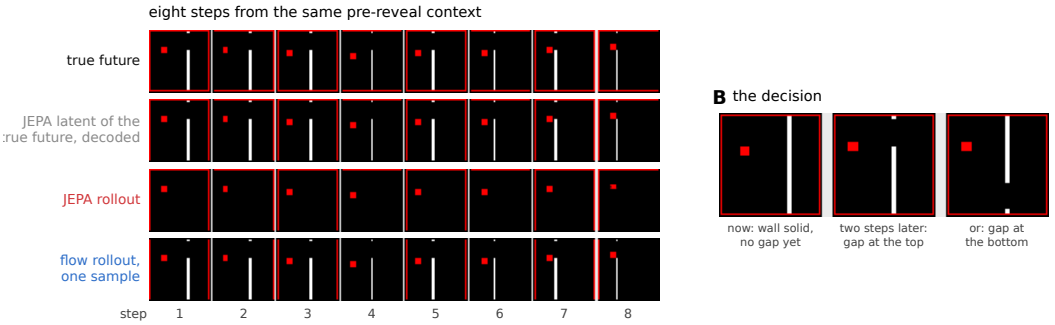


Figure 1: **Encoded walls and imagined walls.** **A**: the true future, decoded real encodings, the original JEPA rollout, and one flow sample at matched future steps from a pre-reveal context (JEPA model seed 0). The decoder renders the wall from real encodings but loses it from predicted encodings; the flow sample retains a wall and commits to a gap. **B**: the gate decision: retreat creates time to respond when the initially hidden gap is revealed. The flow row is a single-checkpoint feasibility reference.

from state information, and exact decodability from prediction-loss scale. They do not explain the empirical failure: neither the geometric diagnostic nor residual sampling identifies a cause.

2 Objective, Task and Models

Objective. The encoder $z_t = f_\theta(x_t) \in \mathbb{R}^D$ and causal predictor are trained jointly with latent history $c_t = z_{t-H+1:t}$ and matching action history $a_{t-H+1:t}$:

$$\mathcal{L}(\theta) = \widehat{\mathbb{E}}_t \| g_\theta(c_t, a_{t-H+1:t}) - z_{t+1} \|^2 + \lambda \, \mathrm{SIGReg}(\{z_t\}). \tag{1}$$

Gradients reach both predictions and targets, without a stop-gradient or teacher. For random unit directions u_p, knots $t_k \in [0, 3]$, quadrature weights ω_k and $w(t) = e^{-t^2/2}$, the empirical regularizer is

$$\mathrm{SIGReg} = \frac{1}{P} \sum_{p,k} \omega_k w(t_k) \left[(\widehat{\mathbb{E}} \cos(t_k \langle z, u_p \rangle) - w(t_k))^2 + (\widehat{\mathbb{E}} \sin(t_k \langle z, u_p \rangle))^2 \right]. \tag{2}$$

It compares projected characteristic functions with a standard Gaussian, constraining the marginal law.

Gate and data. A red 6×6 cube moves in 63×63 RGB frames; four actions move it three pixels left, up, right or down. A white wall three pixels wide enters at $x = 60$ every 22 steps and moves left three pixels per step; its 16-pixel gap is at the top or the bottom with probability $1/2$ and is drawn only once the wall is past $x = 40$. Either gap is five moves from the vertical centre, from which the cube sees the gap two steps before contact, so the surviving strategy is to retreat, wait for the reveal, then move (Figure 1B). Contact with the wall or the frame border ends the episode at reward -1; other rewards are zero and the cap is 128 steps. Training uses 500,000 uniform-random transitions, evaluation 25,000. Random and state-blind alternating actions pass 0.06 and 0.02 walls per episode.

Models. The grid JEPA is Equation 1 with a 2.6M tile-transformer encoder, a $4 \times 4 \times 32$ latent grid ($D = 512$) and an 8.3M factored spatial–temporal predictor with an eight-frame history; its encoder is frozen for all later head, decoder and policy fitting. The flow reference (Lipman et al., 2023; Chen et al., 2024; Hafner et al., 2025) is a reconstruction-trained tokenizer to the same latent shape, a causal latent flow-matching predictor over a 32-frame window, and reward and termination heads trained with it. It is one checkpoint trained on the same data and ten-minute budget, and it differs from the JEPA in architecture, supervision and history length at once.

3 Control Comparison

Recurrent PPO (Schulman et al., 2017; Espeholt et al., 2018; Farebrother et al., 2024) learns either inside a frozen model or in the real gate. In imagination it observes predicted latents and uses learned

reward and termination heads; evaluation always uses real episodes from reset. Real-transition PPO observes the same frozen encoder's latents and receives true rewards and terminations. Both use the same MLP torso design, but real training changes the step budget, rollout starts, collection length and update organization (Appendix C). The true-pixel MLP control shares that torso design, not its input width.

Each JEPA predictor variant has three policy seeds for each of three encoder seeds. The original predictor and a fitted residual predictor, sampled once or averaged over eight samples, give 27 policies in total. The flow reference has six sampled and three mean-of-eight policies on its single checkpoint. Real frozen-latent and decoded-latent CNN controls each have nine runs; true-pixel controls have three per torso. Every policy is evaluated on 256 episodes at its *final* checkpoint. Walls passed per episode is the primary endpoint; discovery means final survival above 50%. Model fits and policy seeds are distinct replication levels.

4 THEORY

The objective does not guarantee useful state or a fixed loss scale for decodable state. These population statements assume explicit function classes and do not describe optimization trajectories.

4.1 A NON-COLLAPSED, CONTROL-INSUFFICIENT GLOBAL OPTIMUM

For a latent law on $\mathbb{R}^D$, define the population regularizer

$$\mathrm{SIGReg}_\infty(\mathcal{L}(z)) = \int_{\mathbb{S}^{D-1}} \int_0^3 w(t) \left[\big(\mathbb{E}\cos(t\langle z,u\rangle) - w(t)\big)^2 + \big(\mathbb{E}\sin(t\langle z,u\rangle)\big)^2 \right] dt\, d\sigma(u), \quad (3)$$

where $w(t) = e^{-t^2/2}$ and σ is uniform on the sphere. This nonnegative functional vanishes for $z \sim \mathcal{N}(0, I_D)$; we use only this direction, and the finite-sample regularizer differs from it by an estimator bias (Appendix A).

Assume **(A1)** observations have form $x_t = R(p_t, b)$, and a measurable map β recovers the episode-constant nuisance b. Under the training distribution, b is independent of the controlled trajectory and actions. For deployment, assume b is independent of the initial controlled state and all exogenous transition randomness. Controlled dynamics and rewards do not otherwise depend on b. Assume **(A2)** a measurable map φ satisfies $\varphi(b) \sim \mathcal{N}(0, I_D)$. Assume **(A3)** the encoder class contains $\varphi \circ \beta$, and the predictor class contains the last-latent copy map. These realizability assumptions are substantive; finite pixel textures need not satisfy them exactly.

Proposition 1 (Texture trap)**.** *Under (A1)–(A3), an encoder and predictor attain zero population prediction loss and zero* SIGReg_∞*. Their latent has covariance I_D and satisfies $I(z_t; p_{1:T}) = 0$ under the training distribution. Any deployed controller observing only these latents, time, and its own actions has the value of a randomized state-blind policy, at most the supremum over such policies.*

Proof. Set $f^\star = \varphi \circ \beta$ and let $g^\star$ copy its last latent. Then $z_t = \varphi(b) = z_{t+1}$, so every prediction residual vanishes. The Gaussian marginal gives covariance I_D and zero population regularizer. Both objective terms are nonnegative, so their sum is globally minimized at zero. Training-distribution independence gives $I(\varphi(b); p_{1:T}) = 0$.

For deployment, draw the nuisance B and the controller's random seed U at reset, independently of the initial state and of the transition randomness; run the original controller in the environment with nuisance B, and let the state-blind policy, with private seed (B, U), feed the constant history $\varphi(B)$ to a copy of that controller and reproduce its actions with U. Coupled to the same initial state and transition randomness, the two policies choose the same first action, and if their actions agree through step t their states, latent histories and seeds agree, so their next actions agree; by induction trajectories, returns and terminations coincide. Nothing here assumes independence between b and states influenced by its induced actions. □

Corollary 1 (An innovation cost for state-resolving encoders)**.** *Let $p_{t+1} = p_t + \mu(a_t) + w_t$ and $\mathcal{H}_t = \sigma(p_{\le t}, b, a_{\le t})$. Write $\Sigma_t = \mathrm{Cov}(w_t \mid \mathcal{H}_t)$ and assume finite second moments for states and encoded observations. Assume observations have the same form $x_t = R(p_t, b)$ with fixed nuisance*

b. Suppose $\|f(R(p,b)) - f(R(p',b))\| \geq \kappa\|p - p'\|$ *for all admissible* p, p', b*, with* $\kappa > 0$*. For every predictor using latent and action history,* $\mathbb{E}\|g(z_{\leq t}, a_{\leq t}) - f(x_{t+1})\|^2 \geq \kappa^2 \mathbb{E}\mathrm{tr}\Sigma_t$ *(Equation 4 in Appendix A). If* $\mathbb{E}\mathrm{tr}\Sigma_t > 0$*, this exceeds the texture encoder's zero prediction loss under (A1)–(A3).*

The bound is the innovation after conditioning on state history and nuisance, so a persistent wind is not automatically innovation, and it covers encoders that preserve every state distance at a uniform scale; Appendix A proves it without a conditional-independence assumption on the wind.

The per-episode-texture cube world illustrates the nuisance mechanism of Proposition 1: the class-token JEPA reaches near-zero prediction loss by encoding texture, while its reward-driven controllers attain 0–2% survival (Appendix B); this finite experiment is not an exact realization of the Gaussian construction.

4.2 The squared-loss target

Lemma 1 (Conditional mean)**.** *For a fixed encoder and data law with* $\mathbb{E}\|z'\|^2 < \infty$*, the unique squared-error minimizer over measurable square-integrable predictors from history* c *and action* a *is* $m(c,a) = \mathbb{E}[z' \mid c, a]$*, up to null sets.*

The orthogonal-decomposition proof is in Appendix A. This unrestricted optimum need not equal a fitted network or an average of finitely many learned samples.

4.3 Decodable information can have arbitrarily small amplitude

Proposition 2 (Small amplitude under an exactly Gaussian marginal)**.** *For every* $\varepsilon > 0$ *and integer* $D \geq 1$*, there is an observation process* $x_t = R(p_t, b)$ *with episode-constant nuisance* b*. Its state* $p_t \in \{0,1\}$ *is a fresh fair bit, drawn independently of* b*, previous states, and previous actions. There is a measurable encoder* f_ε *with* $z_t \sim \mathcal{N}(0, I_D)$ *from which* p_t *and* b *are exactly recoverable almost surely. For the same nuisance, the two state encodings satisfy* $0 < \|f_\varepsilon(R(1,b)) - f_\varepsilon(R(0,b))\| < 2\varepsilon$*. Over all square-integrable measurable predictors of latent and action history through* t*, the optimal next-latent prediction is their midpoint* $m(b)$*, with* $0 < \mathcal{L}_{\mathrm{pred}}(f_\varepsilon, m) \leq \varepsilon^2$*,* $I(m(b); p_{t+1}) = 0$ *and* $\mathrm{SIGReg}_\infty(\mathcal{L}(z_t)) = 0$*. The state is control-relevant: with actions* $a_t \in \{0,1\}$ *and reward* $\mathbf{1}\{a_t = p_t\}$*, latent feedback earns* 1 *per step and every state-blind policy* $1/2$*.*

Proof idea. Cut $\mathbb{R}$ into intervals of width 2ε and let the nuisance pick an interval with its Gaussian mass and a uniform position inside it; let the fair bit decide whether that position lands in the lower or the upper half of the interval's conditional Gaussian mass. The first latent coordinate is then exactly $\mathcal{N}(0,1)$, the bit is read off from the half, the two encodings of the bit lie in one interval, and the optimal predictor is their midpoint, which depends on the nuisance only and costs at most ε^2. Appendix A gives the construction in full. □

The construction is measurable, with no finite-network or robust-decoding guarantee; it averages a fresh innovation and does not by itself explain the deletion of a predictable wall.

5 Results

5.1 Real-transition training outperforms the imagination pipeline

With MLP torsos, real-transition training on the frozen JEPA encoders passes 0.94 walls per episode across nine runs, against 0.27 across the 27 imagined-training runs (Figure 2; Appendix F). This 0.67-wall deficit belongs to the tested imagination-and-training pipeline: dynamics, fitted heads and PPO protocol change together. None of the imagined policies discovers the strategy; their maximum final survival is 3.1%. The real frozen-latent policies also do not discover it, and their wall counts do not establish that they use wall information.

True-pixel MLP policies pass 0.91 walls per episode in three runs without discovery. This control prevents attributing the frozen-latent MLP outcome to missing encoded information; it establishes neither a representational ceiling nor a capacity ceiling for the MLP class. In the flow reference, eight of nine policies pass 0.72–1.18 walls and one passes 3.95. All share one dynamics checkpoint,

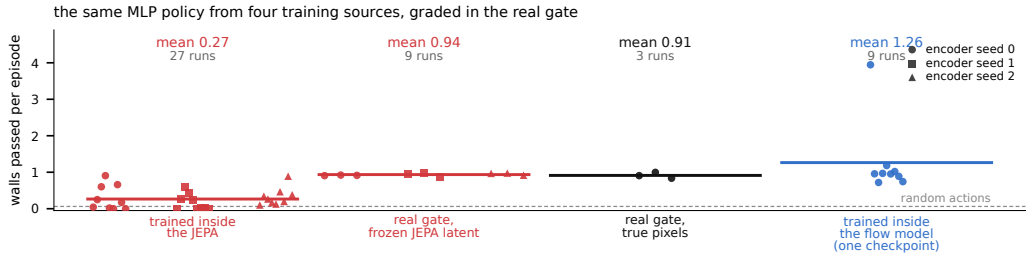


Figure 2: **The primary control contrast.** Final walls passed per real episode, with one dot per policy and a horizontal mean marker. All groups use MLP torsos: imagined JEPA (27 policies), real frozen latent (9), real pixels (3), and imagined flow (9, one checkpoint). Marker shape identifies the JEPA encoder seed; the JEPA group pools its three predictor variants (original, residual sample, residual mean of eight) and the flow group its two sampling modes. The real-transition advantage measures the complete pipeline contrast, including the PPO protocol.

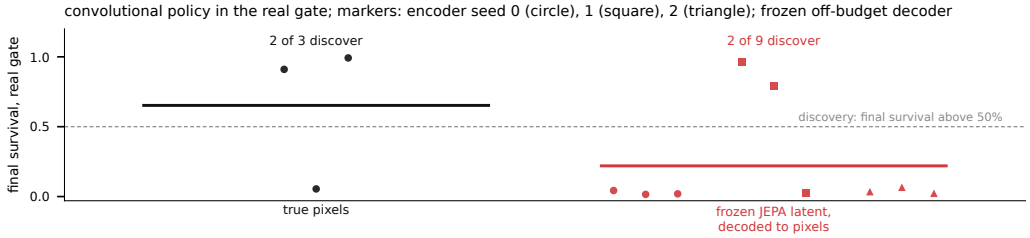


Figure 3: **Useful information remains encoded.** Final survival for true-pixel CNN policies and CNN policies acting on decoded frozen latents, with marker shape identifying the encoder seed. Each dot is a policy run; the dashed line marks the discovery criterion, survival above 50%. The decoded policies discover the strategy in $2/9$ runs, both on encoder 1, versus $2/3$ true-pixel runs. Decoder supervision is off-budget. These are final endpoints, not training curves.

so policy replication does not establish family-level superiority. Training the gate JEPA for twice and four times the steps, $17{,}000$ and $34{,}000$, leaves in-model PPO at 0–3% survival (model seed 0, policy seed 0, each predictor variant).

5.2 Encoded information can support discovery

A CNN trained in the real gate on decoded frozen latents discovers the strategy in $2/9$ runs, both with encoder 1 (Figure 3). The corresponding true-pixel CNN discovers it in $2/3$ runs. Decoder fitting adds reconstruction supervision outside the world-model budget. The result establishes useful information in at least one encoder under this read-out and learning recipe, without asserting that every encoder preserves all task-relevant information.

Decoded real encodings retain the wall, but the original predicted rollouts lose it across all three encoder seeds: by the second predicted step, decoded wall intensity is at most 0.02 (Appendix F). Figure 1 shows the contrast with a flow sample that retains the wall and commits to a gap. A pixelwise mixture of the two gap futures would preserve shared wall pixels; the observation differs from that explanation, but does not exclude averaging effects in the learned latent model.

The fitted heads are a named competing explanation: on encoder 0, termination AUC is 0.96 for border deaths and 0.65 for wall deaths, while predicted rewards are -0.89 and -0.05, respectively, against the true -1; these are the signals on which the in-model policy trains.

5.3 Scale and sampling bound the interpretation

Counterfactual wall distance is smaller than total teacher-forced prediction error (Figure 4). On 64 pre-reveal contexts for encoder 0, the first-step squared distance between wall-present and wall-absent encodings is 0.035, against a one-step residual of 0.145. The distance measures sensitivity

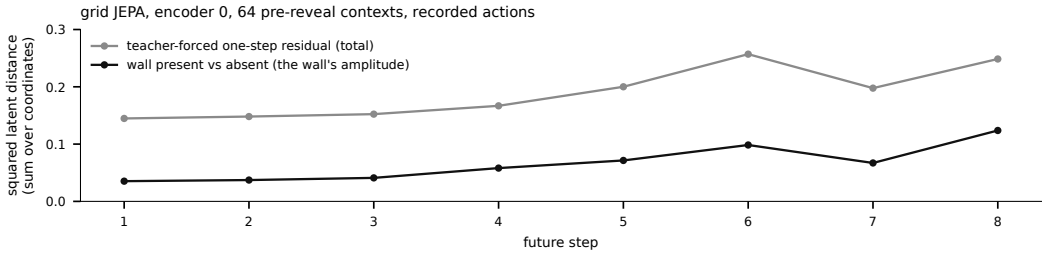


Figure 4: **A descriptive scale comparison.** Squared latent wall distance and total teacher-forced one-step residual versus future step, averaged over the 64 pre-reveal contexts for JEPA model seed 0. Wall distance compares real encoded futures with and without the wall, under matched actions. Its position below total prediction error does not identify a cause of wall deletion. No cross-model scale comparison is implied.

to wall removal, whereas the residual includes all predicted features. Their ordering is descriptive evidence, not a test that small amplitude causes wall deletion or control failure. Raw distances across independently learned latent spaces are not comparable information scales.

Sampling the fitted residual predictor does not rescue control: one sample and the mean of eight each give 0.26 walls per episode, against 0.27 for the original predictor, with no discoveries. The residual fitting protocol is described in Appendix C. These tests neither implicate nor acquit the mean. They test a fitted distribution with shared heads and training protocol, not an exact conditional mean or every distributional predictor.

6 Related work

Joint-embedding prediction learns targets in representation space (LeCun, 2022; Assran et al., 2023; Bardes et al., 2024). LeWM jointly trains an encoder and an action-conditioned predictor with squared loss and SIGReg (Maes et al., 2026; Balestriero & LeCun, 2025); other systems plan on frozen features (Assran et al., 2025; Zhou et al., 2024; Sobal et al., 2025). Anti-collapse methods constrain learning dynamics or latent statistics (Grill et al., 2020; Chen & He, 2021; Zbontar et al., 2021; Bardes et al., 2022; Jing et al., 2022). Our population counterexample has full covariance rank yet no controlled-state information.

Predictable backgrounds can displace moving objects in joint embeddings (Sobal et al., 2022). Analyses of self-predictive abstractions (Tang et al., 2023; Ni et al., 2024) and prediction–control mismatch (Lambert et al., 2020) motivate stronger criteria than prediction accuracy. Reward-, transition- and value-based abstractions supply such criteria (Ferns et al., 2004; Zhang et al., 2021; Grimm et al., 2020; Farahmand et al., 2017). Our contribution is an exactly Gaussian nuisance-only optimum, a decodable small-amplitude construction, and a measured pipeline deficit alongside successful control from decoded latents.

Imagination-based policy learning (Sutton, 1991; Ha & Schmidhuber, 2018; Hafner et al., 2020) can use stochastic dynamics (Hafner et al., 2019; 2023; Alonso et al., 2024; Hafner et al., 2025). Our flow reference demonstrates feasibility for this task and budget; it does not isolate stochasticity or compare model families. Likewise, the residual-sampling null is specific to the fitted predictor.

7 DISCUSSION AND LIMITATIONS

Real-transition training outperforms the tested imagination-and-training pipeline on the same encoders, and at least one frozen encoder supports discovery through a decoder and CNN. The population objective nevertheless admits control-insufficient Gaussian optima and decodable state at arbitrarily small loss scale. These existence results do not explain optimization or finite-network behaviour; the fresh-bit construction does not explain predictable-wall deletion. The scale diagnostic is descriptive, and the sampling null neither implicates nor acquits averaging.

Empirical scope is one objective, a small gate task, a grid architecture and the reported budgets. Predictor, head and PPO-protocol effects remain unresolved. MLP results bound neither representation nor policy capacity; decoded discoveries require off-budget supervision and are confined to encoder 1. The flow reference is one checkpoint with different supervision and history, not independent model replication. Neither non-collapse nor decodability certifies control performance in this pipeline.

REPRODUCIBILITY STATEMENT

Appendices C–F document fitting, evaluation and final-checkpoint outcomes, distinguishing model and policy replication. The supplied ledger supports the reported aggregates. Complete configurations, code and data generation will accompany an anonymous artifact; the ledger alone does not specify every fitting setting.

STATEMENT ON THE USE OF AI

Language models were used to draft and edit prose, to check the proofs for gaps, to write analysis and plotting scripts, and to act as an adversarial reviewer of the manuscript. The authors are responsible for the experiments, numerical checks, proofs and claims.

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A PROOFS

Proposition 1 is proved in Section 4.

Corollary 1.

Proof. Put $Z = f(x_{t+1})$ and $\mathcal{C}_t = \sigma(z_{\le t}, a_{\le t})$. Because observations and latents are measurable functions of state and nuisance, $\mathcal{C}_t \subseteq \mathcal{H}_t$. Any predictor using a shorter window is also $\mathcal{C}_t$-measurable. Conditional least squares and the tower property give

$$\begin{aligned}\mathbb{E}\|g - Z\|^2 &\ge \mathbb{E}\|Z - \mathbb{E}[Z \mid \mathcal{C}_t]\|^2 \\ &= \mathbb{E}\|Z - \mathbb{E}[Z \mid \mathcal{H}_t]\|^2 + \mathbb{E}\|\mathbb{E}[Z \mid \mathcal{H}_t] - \mathbb{E}[Z \mid \mathcal{C}_t]\|^2 \\ &\ge \mathbb{E}\mathrm{trCov}(Z \mid \mathcal{H}_t).\end{aligned}$$

The cross term vanishes because its first factor has conditional mean zero given $\mathcal{H}_t$. Take conditionally independent copies P, P' of p_{t+1} given $\mathcal{H}_t$. Such copies exist for Euclidean state spaces, using their regular conditional distributions. The nuisance b and current state p_t are fixed under this conditioning. The conditional independent-copy variance identity yields

$$\begin{aligned}\mathrm{trCov}(Z \mid \mathcal{H}_t) &= \tfrac{1}{2}\mathbb{E}[\|f(R(P,b)) - f(R(P',b))\|^2 \mid \mathcal{H}_t] \\ &\ge \tfrac{\kappa^2}{2}\mathbb{E}[\|P - P'\|^2 \mid \mathcal{H}_t] \\ &= \kappa^2 \mathrm{trCov}(p_{t+1} \mid \mathcal{H}_t) = \kappa^2 \mathrm{tr}\Sigma_t.\end{aligned}$$

The final equality follows because $p_t + \mu(a_t)$ is $\mathcal{H}_t$-measurable. Taking expectations proves Equation 4. Averaging these inequalities gives the corresponding bound for a multi-time prediction objective. Proposition 1 supplies the zero-loss comparator when its assumptions also hold. □

Lemma 1.

Proof. Let $m = \mathbb{E}[z' \mid c, a]$. For every square-integrable measurable $g(c, a)$, expand $g - z' = (g - m) + (m - z')$. The cross term has expectation zero since $\mathbb{E}[m - z' \mid c, a] = 0$. Hence

$$\mathbb{E}\|g - z'\|^2 = \mathbb{E}\|g - m\|^2 + \mathbb{E}\mathrm{trCov}(z' \mid c, a).$$

The second term does not depend on g. The first is nonnegative and vanishes exactly when $g = m$ almost surely. □

The innovation bound. Corollary 1 states

$$\mathbb{E}\|g(z_{\le t}, a_{\le t}) - f(x_{t+1})\|^2 \ge \kappa^2 \mathbb{E}\mathrm{tr}\Sigma_t. \tag{4}$$

The finite-sample regularizer. The finite-sample regularizer differs from Equation 3. For independent identically distributed samples with characteristic function ϕ and empirical estimate $\widehat{\phi}_n$,

$$\mathbb{E}|\widehat{\phi}_n(t) - w(t)|^2 = |\phi(t) - w(t)|^2 + \frac{1 - |\phi(t)|^2}{n}. \tag{5}$$

Indeed, $\mathbb{E}\widehat{\phi}_n = \phi$ and $\mathbb{E}|\widehat{\phi}_n - \phi|^2 = (1 - |\phi|^2)/n$, since each complex exponential has modulus one; expanding around ϕ gives the identity. At a Gaussian law, the residual equals $(1 - e^{-t^2})/n$. This is estimator bias, not an optimization floor. Changing the law changes both terms; dependent temporal samples introduce additional covariance terms. Finite projections and knots also do not characterize Gaussianity.

Proposition 2.

Proof. Partition $\mathbb{R}$ into $I_j = [2\varepsilon j, 2\varepsilon(j+1))$, $j \in \mathbb{Z}$. Let $q_j = \Pr(G \in I_j)$ for $G \sim \mathcal{N}(0,1)$. Every q_j is positive and their sum is one. Let F_j be the conditional Gaussian distribution function on I_j, with inverse quantile function Q_j on $(0,1)$. The conditional CDF is continuous and strictly increasing on the interval interior, so its inverse is measurable and takes values in that interior.

Draw J with probabilities q_j, $U \sim \mathrm{Uniform}(0,1)$, and $V \sim \mathcal{N}(0, I_{D-1})$, independently. Set $b = (J, U, V)$; for $D = 1$, omit V. Take $R(p, b) = (p, b)$ and define

$$f_\varepsilon(R(p,b)) = \big(Q_J((U+p)/2), V\big). \tag{6}$$

The mixture $(U + p_t)/2$ is uniform on $(0, 1)$, since p_t is an independent fair bit. Conditional on $J = j$, the first coordinate therefore has the Gaussian law restricted to I_j. Mixing with weights q_j recovers $\mathcal{N}(0, 1)$. Independence of V gives the joint law $\mathcal{N}(0, I_D)$ and zero population regularizer.

Given the first coordinate y, its interval determines J almost surely. Write $v = F_J(y)$. Then $p_t = \mathbf{1}\{v > 1/2\}$ and $U = 2v - p_t$ almost surely; the other coordinates recover V. Thus the latent recovers both state and nuisance. For fixed b, the two first coordinates are distinct and lie inside the same interval of width 2ε. All remaining coordinates agree, proving the distance bound.

Latent history reveals b, while the next bit remains independent and fair. The next latent consequently has two equiprobable values, say $z^{(0)}(b)$ and $z^{(1)}(b)$. Lemma 1 gives $m(b) = (z^{(0)}(b) + z^{(1)}(b))/2$. Its conditional squared error is $\|z^{(1)}(b) - z^{(0)}(b)\|^2/4$, strictly positive and bounded by ε^2. The midpoint is square-integrable by Jensen's inequality, since both state encodings have finite second moments. Taking expectations proves the loss bound, including strict positivity since the conditional error is positive almost surely. The midpoint depends only on b, so it is independent of p_{t+1}. Finally, decoding the current bit permits $a_t = p_t$. A state-blind action is independent of the current fair bit, giving success probability $1/2$. □

B THE CUBE WORLDS

The cube worlds remove the gate wall and cross black, per-frame random and per-episode random backgrounds with still or hidden-wind dynamics. The class-token JEPA's near-zero prediction loss on per-episode texture is consistent with retaining predictable nuisance rather than controlled state, as in Proposition 1; finite textures do not meet its exact Gaussian realizability assumption. Table 1 retains the original survival outcomes for context. Still-world survival need not require state feedback. These supplementary worlds do not extend the scope of the gate comparison or identify its failure mechanism.

Table 1: Survival (percent) in the supplementary cube worlds. "LeWM" denotes the class-token JEPA; "grid" denotes the grid JEPA; "goal" and "heads" use latent-goal and learned-return search, respectively. PPO columns use the final saved policies; "px" and "z" identify pixel and latent observations. Still-world survival does not certify state feedback.

Background	Black		Per-frame noise		Per-episode noise	
Dynamics	Still	Wind	Still	Wind	Still	Wind
random	20	0	20	0	20	0
centre	100	99	100	99	100	99
LeWM goal	97	64	5	0	80	0
LeWM heads	100	60	9	0	2	0
LeWM PPO	100	99	0	0	0	0
grid goal	88	1	92	0	100	77
grid heads	100	5	100	0	85	8
grid PPO	100	94	100	0	2	0
flow PPO px	100	94	98	97	98	95
flow PPO z	100	96	99	1	51	1
PPO real	100	98	100	96	100	78

C HYPERPARAMETERS AND COMPUTE

Each world supplies 500,000 uniform-random transitions. Model training uses bfloat16 compute, float32 parameters, and one A100. A 500-step calibration estimates the training length fitting 575 seconds including compilation; the grid gate model trains for 8,500 steps (17,000 and 34,000 in the budget sentence of Section 5), the class-token cube models for 23,750–24,000. Model seeds 0, 1, 2 of the grid JEPA differ only in the initialisation and data-order seed.

JEPA settings. Both arms use batch 128 and AdamW with learning rate and weight decay 10^{-3}, 10% warm-up followed by cosine decay to zero, and gradient clipping at 1.0. SIGReg uses weight 0.09, 17 knots on $[0, 3]$, and 1024 random projections. Neither arm uses a stop-gradient, teacher, decoder, or reward during world-model fitting.

	class-token JEPA (`lewm6`)	grid JEPA (`tile_factored`)
encoder	ViT, 7×7 patches	tile transformer + strided conv
width / depth / heads	192 / 6 / 3	128 / 6 / 8
encoder MLP	768	$4\times$ width
read-out	class token	4×4 grid, 32 channels
latent dimension	192	512
projector	linear-BN-GELU-linear, 2048	same, hidden 512
predictor	AdaLN-zero autoregressive	spatial then causal temporal
predictor depth / heads / head dim	6 / 16 / 64	6 / 8 / 32
predictor MLP / width	2048 / latent	$2\times$ width / 256
history	3 frames	8 frames
dropout	0.1	0
parameters	15.3M	2.6M + 8.3M

Flow settings. The tokenizer uses 9,000 steps, batch 256, and learning rate 9×10^{-4}; dynamics 16,000 steps, batch 64, the same learning rate, width 256, six layers, eight spatial and temporal attention heads, a 32-frame context, independent per-frame noise levels, a shortcut loss of weight 0.25 every fourth step after the first quarter of training, history dropout 0.25, two history frames resampled every eight steps after half of training, and a temporal loss weight of 0.25. The symlog reward head has 255 bins over $[-20, 20]$; reward and termination losses have weight 0.1 and enter the trunk at full strength for half of training, then at factor 0.1. Inference uses four function evaluations per frame. We corrected terminal-transition masking in the recipe's reward loss (Appendix G).

JEPA read-outs. Post-hoc heads use the frozen encoder and 80,000 frames from a different behaviour-policy seed, with the last 200 time rows held out. The pixel decoder uses 8,000 steps, batch 256, learning rate 10^{-3}, width 256, six residual blocks; it supports the figures and the decoded-latent control, in which its frozen output, clipped to $[0, 255]$, replaces the pixels of the real-gate PPO; no true pixel enters that policy after the encoder. Reward, termination, and behaviour-value heads are two-layer perceptrons of width 512 on the latent and the one-hot action, 6,000 steps, batch 1024, learning rate 10^{-3}, Monte Carlo value targets at discount 0.99, value loss weight 0.1. The residual predictor models $z_{t+1} - m$ with $m = g(z_{t-7:t}, a_{t-7:t})$ the original grid prediction, conditioned on m, z_t, and the action: three residual blocks of width 1024, 8,000 steps, batch 2048, 74 seconds, four function evaluations per sample. On 20,000 held-out windows the original predictor's squared error is 3.07×10^{-4}, one residual sample 2.05×10^{-4}, the mean of eight 1.32×10^{-4}.

Policy optimisation. PPO uses discount 0.99, GAE parameter 0.9, clipping 0.2, value coefficient 0.5, entropy coefficient 0.01, AdamW at 2.5×10^{-4} with weight decay 0.01, four epochs per batch, and gradient clipping at 1.0. Pixels pass through a three-block Impala torso, latents through a two-layer perceptron; a GRU receives that embedding and the previous action; the critic uses 64 categorical bins with histogram-Gaussian targets. Real-environment PPO uses 128 environments and 64 steps per iteration with four minibatch groups, truncated backpropagation over 16 steps, and 1.5M environment steps. In-model PPO uses 150 updates of 512 trajectories of 16 steps (1.2M imagined steps) with minibatch size 128; half of the starts are resets, half dataset states with their real eight-frame context. All reported policy rows use final checkpoints.

Compute. Logged training times on a shared A100: grid JEPA training 573 s (8,500 steps; 1,074 and 3,802 s for the 17,000- and 34,000-step fits); probe and heads 3 min; residual predictor 74 s; PPO inside the JEPA 2.0–4.3 min per run; PPO inside the flow model 5–6 min per run at four function evaluations, 20–26 min with the mean of eight; PPO in the real environment 3.7–7.7 min on pixels with the convolutional torso and 51–70 s with the MLP torso, 76–78 s on the frozen latent, 4.7–4.8 min on the decoded latent with the convolutional torso and 1.8–2.9 min with the MLP torso; the latent diagnostic of Figure 4: 2 min.

D EVALUATION PROTOCOL

Every gate-policy endpoint uses the final checkpoint and 256 real episodes from reset, with the 128-step cap. Survival is the fraction reaching the cap. Walls passed counts walls leaving the frame while the agent remains alive. Discovery is final survival strictly above 50%. The same evaluation harness shares environment random keys across arms. Policy seeds are separate runs conditional on a fitted model; independently fitted encoders provide a different replication level. The flow model has only one checkpoint. Reported means weight policy runs equally; standard deviations describe variation across those runs, not uncertainty across model families.

Latent diagnostic. For each encoder, the diagnostic uses 64 real contexts with the wall still solid at $x \in \{42, 45\}$. Reconstructed simulator states generate recorded futures and matched wall-free futures under the recorded actions. The recorded branch survives the diagnostic horizon by construction. At each step, wall amplitude is the squared Euclidean distance between encodings of the wall-present and wall-absent frames. The teacher-forced residual uses real encoded history and is a total squared error, summed over latent coordinates. The wall coordinate of a prediction is its projection onto the wall-present minus wall-absent latent difference, with endpoints 0 and 1. Decoded wall intensity averages the minimum colour channel on shared wall pixels, excluding either possible gap band. These measurements compare features within one encoder; raw distances across models are not an information metric.

E LATENT-POLICY TORSO CONTROLS

The MLP on decoded latents passes 0.92 walls per episode with no discoveries in six runs (Table 2), using encoders 0 and 1. The decoded CNN control instead uses all three encoders, with three policies per encoder. Both of its discoveries use encoder 1. The decoder is frozen and reconstruction-supervised; success therefore establishes useful encoded information under this added read-out, not sufficiency of the unsupervised training objective. Failure of the true-pixel MLP recipe establishes no upper bound on MLP capacity.

Table 2: Real-gate observation and torso controls: final walls per episode and survival, mean $\pm$ standard deviation over policies; discovery means final survival above 50%. All policies use $1{,}499{,}136$ real transitions.

observation of the real-gate PPO	runs	walls passed	survival (%)	discovered
pixels, convolutional torso	3	2.91±1.65	65±52	2/3
pixels, MLP torso	3	0.91±0.08	4±2	0/3
frozen JEPA latent, MLP torso	9	0.94±0.04	4±1	0/9
decoded JEPA latent, convolutional torso	9	1.57±1.26	22±37	2/9
decoded JEPA latent, MLP torso	6	0.92±0.08	3±2	0/6

F CONTROL OUTCOMES AND LATENT DIAGNOSTICS

Table 3 gives aggregate final outcomes; Table 4 lists every imagined JEPA policy. The pooled imagined JEPA mean weights all original, sampled-residual and mean-residual runs equally. The flow rows distinguish six sampled policies from three policies using a mean of eight, all sharing the same model fit. They do not estimate variation across independent flow-model fits. Tables 5 and 6 retain the encoder and geometry diagnostics supporting the gate observations.

G TERMINAL REWARDS IN THE FLOW RECIPE

The original recipe masks transitions entering reset frames out of reward training. Here the only nonzero reward occurs on terminal transitions, so that mask removes the reward signal. The corrected head predicts the next reward from the current frame summary and action with terminal transitions retained in its loss. The reported flow-policy outcomes use this corrected convention. The single-checkpoint reference therefore denotes the complete tokenizer, dynamics, reward and termination package, rather than an isolated change in transition stochasticity.

Table 3: Final control outcomes in the real gate, mean ± standard deviation over policy runs. Imagined training uses 1,228,800 transitions per policy; real training uses 1,499,136. Model-fit counts and policy-run counts identify distinct replication levels.

controller	model seeds	policies	walls passed	survival (%)	discovered
grid JEPA, original predictor	3	9	0.27±0.22	0±1	0/9
grid JEPA, residual head, mean of 8	3	9	0.26±0.30	1±1	0/9
grid JEPA, residual head, one sample	3	9	0.26±0.32	1±1	0/9
flow model, mean of 8 samples	1	3	0.88±0.14	3±1	0/3
flow model, one sample	1	6	1.45±1.23	19±39	1/6
random	–	–	0.06	0	–
state-blind alternating	–	–	0.02	0	–
PPO in the real gate, pixels, CNN torso	–	3	2.91±1.65	65±52	2/3
PPO in the real gate, pixels, MLP torso (latent policy class)	–	3	0.91±0.08	4±2	0/3
PPO in the real gate, frozen JEPA latent, MLP torso	3	9	0.94±0.04	4±1	0/9
PPO in the real gate, decoded JEPA latent, CNN torso	3	9	1.57±1.26	22±37	2/9

Table 4: Every imagined JEPA policy at its final checkpoint: survival (percent) and walls per episode by encoder seed, predictor variant and PPO seed. Each cell uses 256 real episodes.

		survival (%)			walls passed		
predictor	model seed	PPO 0	PPO 1	PPO 2	PPO 0	PPO 1	PPO 2
original predictor	0	0	0	1	0.04	0.25	0.60
	1	0	0	1	0.00	0.26	0.59
	2	0	1	0	0.10	0.34	0.27
residual head, mean of 8	0	3	0	0	0.91	0.02	0.00
	1	2	1	0	0.43	0.25	0.00
	2	0	0	1	0.17	0.13	0.46
residual head, one sample	0	2	1	0	0.66	0.17	0.00
	1	0	0	0	0.02	0.02	0.00
	2	0	3	1	0.20	0.89	0.38

Table 5: Grid-JEPA diagnostics by encoder seed. Wall amplitude and total teacher-forced residual are squared distances at future steps 1 and 8; decoded intensity uses steps 1 and 2. The wall coordinate has endpoints 0 (wall absent) and 1 (true encoding). Head AUC compares each death cause with alive transitions; PPO wall counts pool predictor variants.

JEPA encoder seed	0	1	2
Wall amplitude, step 1	0.035	0.028	0.088
Wall amplitude, step 8	0.124	0.082	0.193
Total residual, step 1	0.145	0.130	0.156
Total residual, step 8	0.249	0.228	0.279
Wall coordinate, step 1	0.39	0.45	0.66
Decoded wall, step 1	0.00	0.00	0.68
Decoded wall, step 2	0.00	0.00	0.02
Border-death AUC	0.96	0.96	0.88
Wall-death AUC	0.65	0.65	0.76
Imagined PPO, walls	0.29±0.34	0.17±0.22	0.33±0.24

Table 6: JEPA geometry and decoded wall intensity for model seed 0, averaged over 64 pre-reveal contexts. Distances use matched actions and the same encoder; total residuals include all latent coordinates. Wall intensity is measured on shared wall pixels.

squared latent distance, rollout step	1	3	5	8
wall present vs absent (encoded true futures)	0.035	0.041	0.071	0.124
predictor's one-step residual (teacher forced, total)	0.145	0.152	0.200	0.249
one-step residual along the wall direction	0.015	0.021	0.048	0.104
one-step residual orthogonal to it	0.130	0.132	0.152	0.145
one-step prediction: mean wall coordinate (0 = no wall, 1 = true latent)	0.39	0.33	0.29	0.21
grid JEPA, encoded true future	1.00	1.00	1.00	1.00
grid JEPA, original rollout	0.00	0.00	0.00	0.00